OAP200V

# Object-oriented analysis, design and programming

## 00 – Running example

# The running example – Cinema Manager

# Problem Description: Cinema Manager

- **Context** - Modern cinemas require a flexible and efficient software system to manage daily operations, including film scheduling, customer reservations, ticketing, and reporting. The system must support multiple cinema locations, a dynamic movie catalog, and customer service operations performed by staff.
- **Problem Statement** - The management of a cinema chain seeks to replace its current manual and fragmented systems with a unified desktop application. This application should allow cinema staff to manage movie screenings, customer reservations, and ticket sales efficiently. The system must support multiple branches and provide real-time access to scheduling, film data, and customer information.

# Cinema manager

- Stakeholders
  - Cinema staff: Responsible for managing screenings, reservations, and customer data.
  - Customers: Interact with the system indirectly through staff for ticket reservations and purchases.
  - Managers: Require access to reports and statistics such as attendance and revenue.

# Cinema manager

- System Goals
  - Manage a catalog of movies with associated genres, languages, durations, and descriptions.
  - Schedule movie screenings across various cinema halls and locations.
  - Register customers and manage their reservations.
  - Handle seat selection and ticketing for individual screenings.
  - Provide accurate and real-time movie and screening information.
  - Generate operational reports including upcoming screenings, sold-out shows, revenue and attendance statistics.

# Cinema manager

- System Environment
  - Desktop application developed using Java (UI using JavaFX)
  - MySQL relational database backend
  - JDBC for database connectivity
  - User interface tailored for staff use

# Use cases

- UC1: Manage Movies Add, edit, and remove movie entries from the catalogue (title, duration, genre, language, etc.).
- UC2: Schedule Screenings Create and update scheduled showings for specific movies, times, and halls.
- UC3: Register Customer Add a new customer to the system with contact details.
- UC4: Make Reservation Reserve a ticket for a specific screening on behalf of a customer.
- UC5: Generate Reports Produce statistics such as attendance rates, most popular films, etc. (Manager only).

# Use case diagram



# Functional requirements

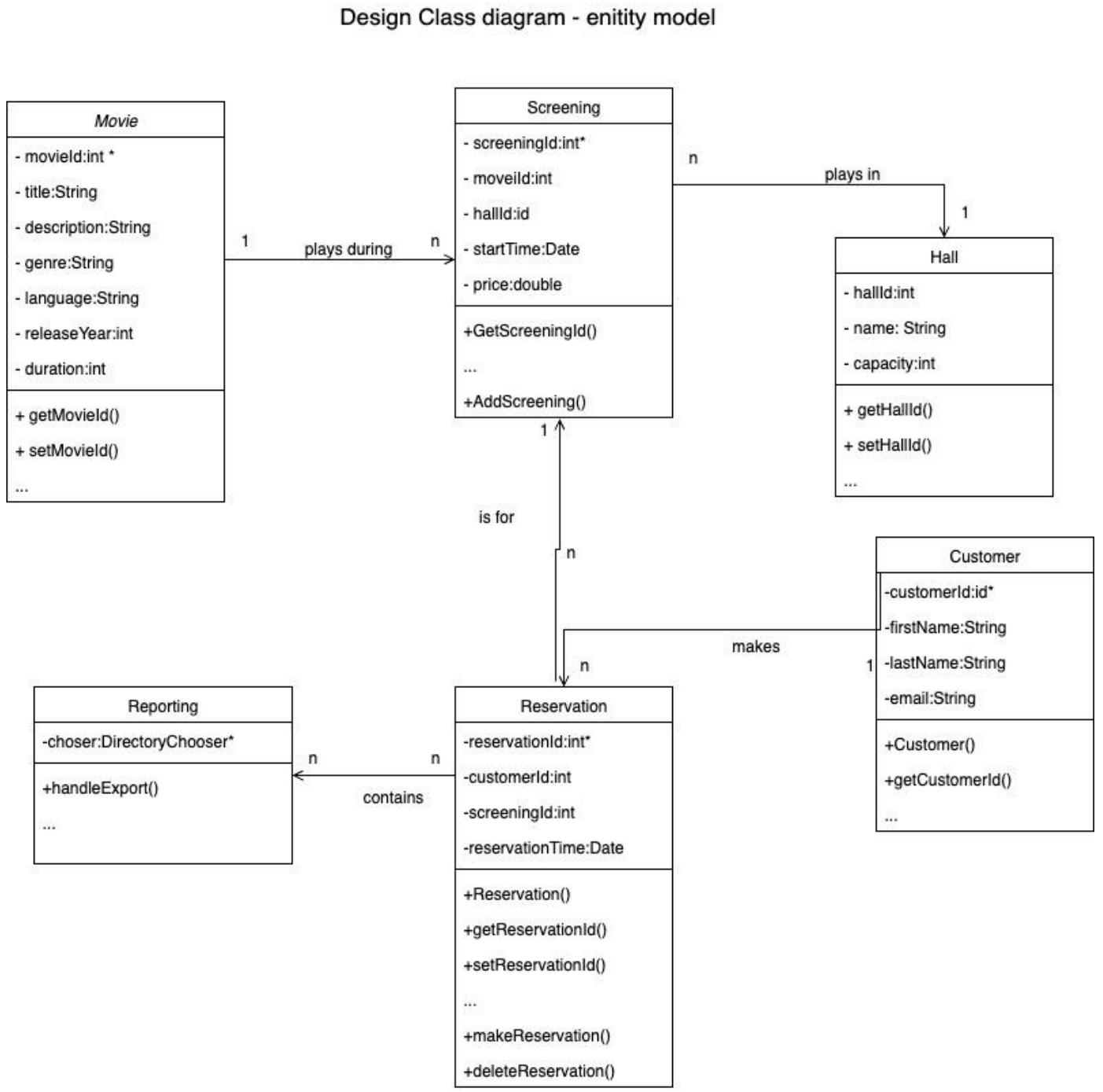
| ID | Requirement | Priority |
|---|---|---|
| FR1 | The system shall allow staff to create, edit, and delete movie entries. | High |
| FR2 | The system shall store metadata for each movie (title, description, genre, language, duration). | High |
| FR3 | The system shall allow scheduling of screenings, specifying the movie, time, and hall. | High |
| FR4 | The system shall prevent overlapping screenings in the same hall. | High |
| FR5 | The system shall allow staff to register customers and store their contact information. | High |
| FR6 | The system shall enable movie reservation and associate it with customers. | High |
| FR7 | The system shall display a searchable and filterable list of upcoming screenings. | Medium |
| FR8 | The system shall allow managers to view summary reports (e.g., movie reservations, most-viewed movies). | Medium |

# Non-functional requirements

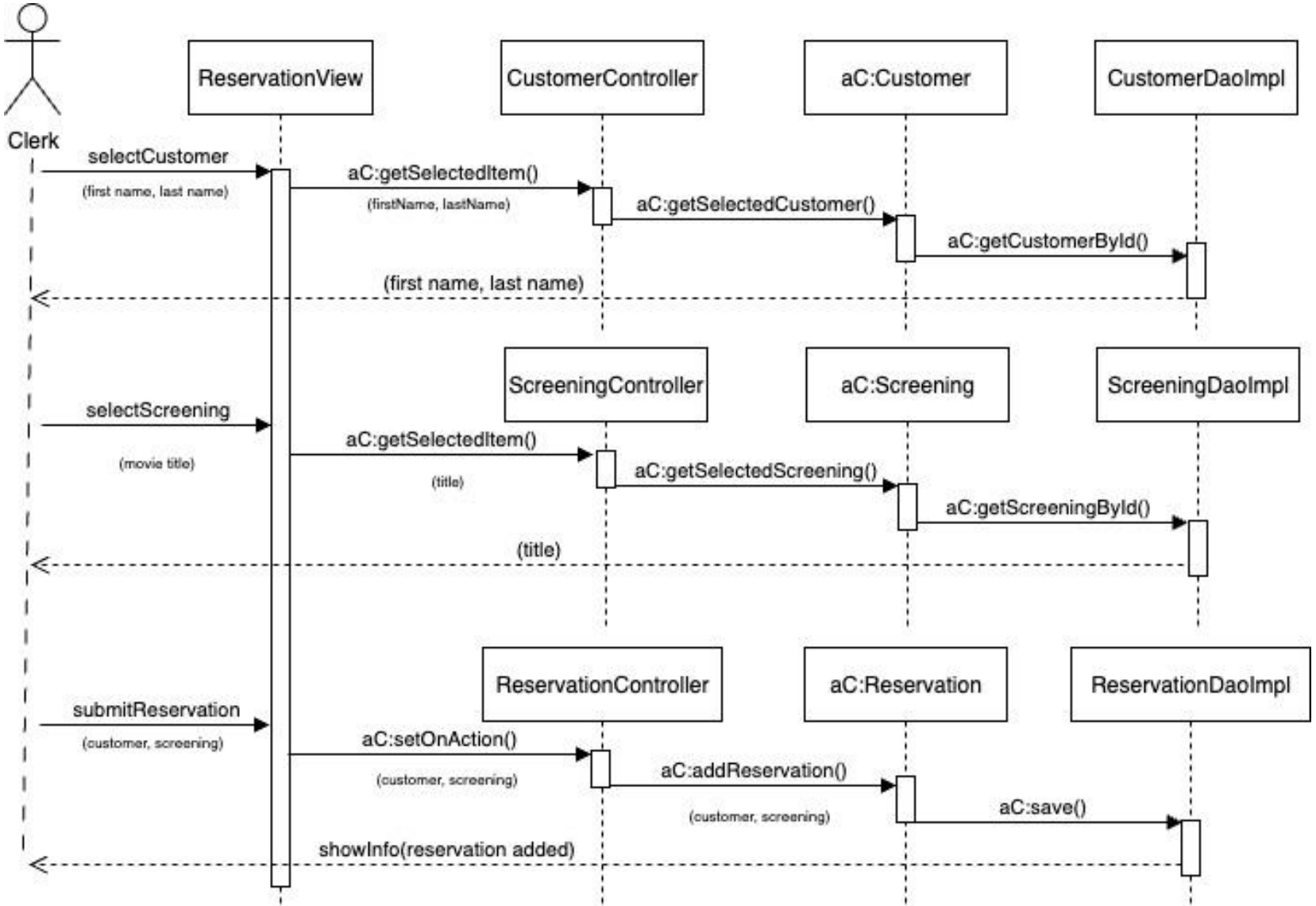
| ID | Requirement | Category | Priority |
|---|---|---|---|
| NFR1 | The application shall be developed in Java using JavaFX for the user interface. | Constraint (Design) | Must |
| NFR2 | The system shall store and retrieve data using a MySQL database via JDBC. | Constraint (Implementation) | Must |
| NFR3 | The application shall provide a responsive and intuitive user interface for staff. | Usability | Should |
| NFR4 | All inputs shall be validated for completeness and correctness. | Accuracy | Must |
| NFR5 | The system shall provide acceptable performance for up to 5 concurrent users. | Performance | Should |

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# Class diagram

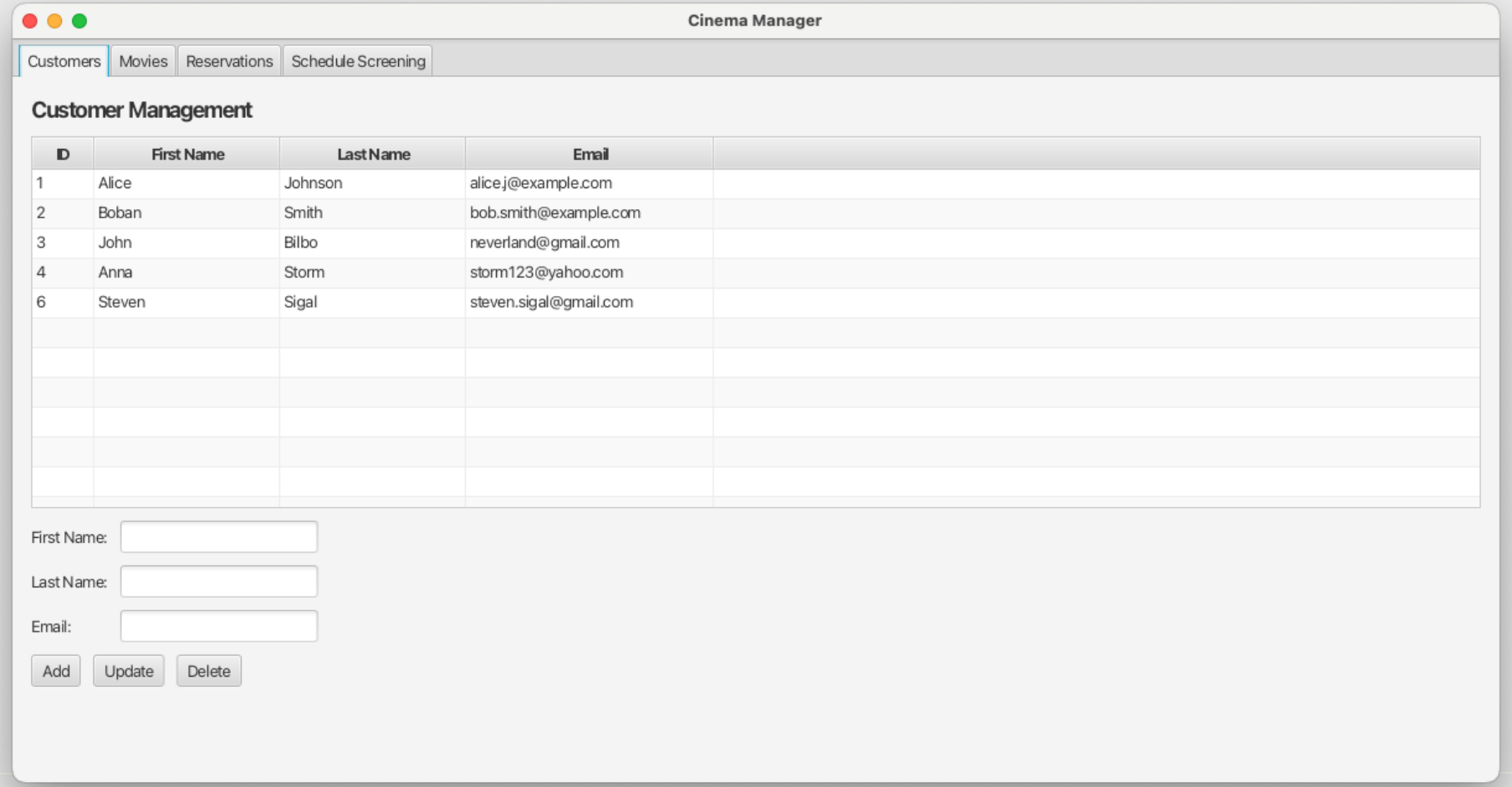


# Sequence diagram

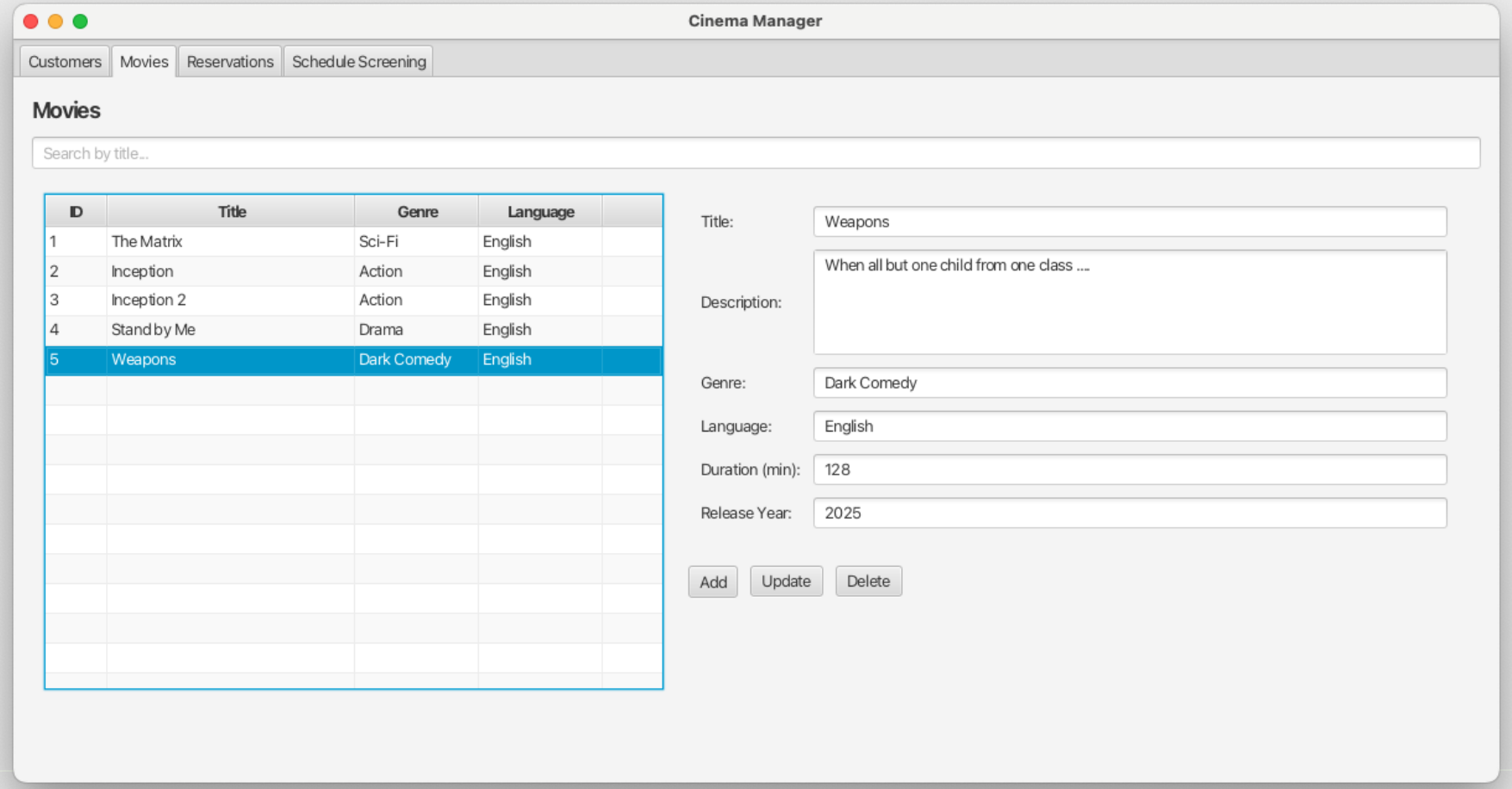


# Implemented app

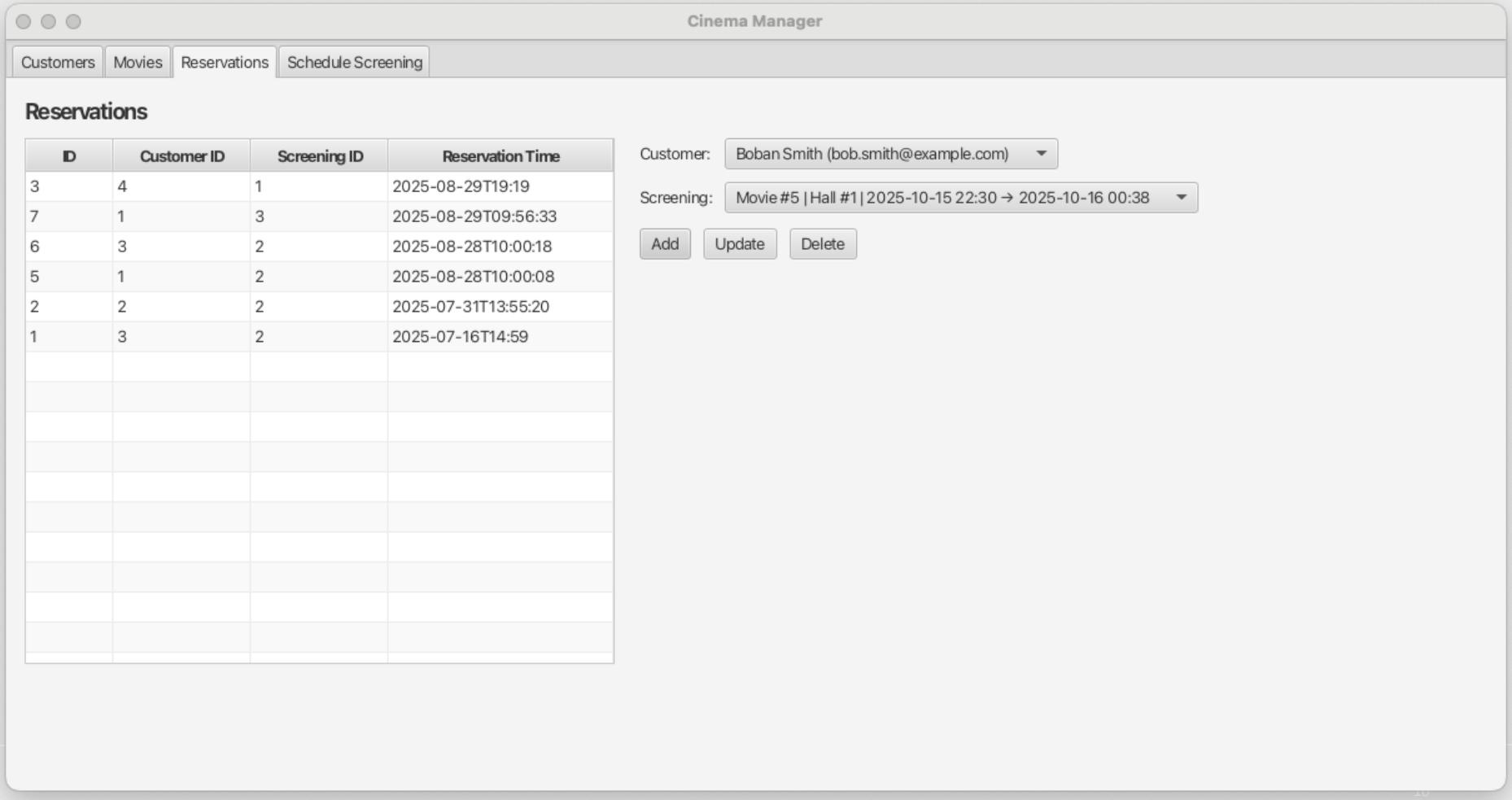
# Customers view



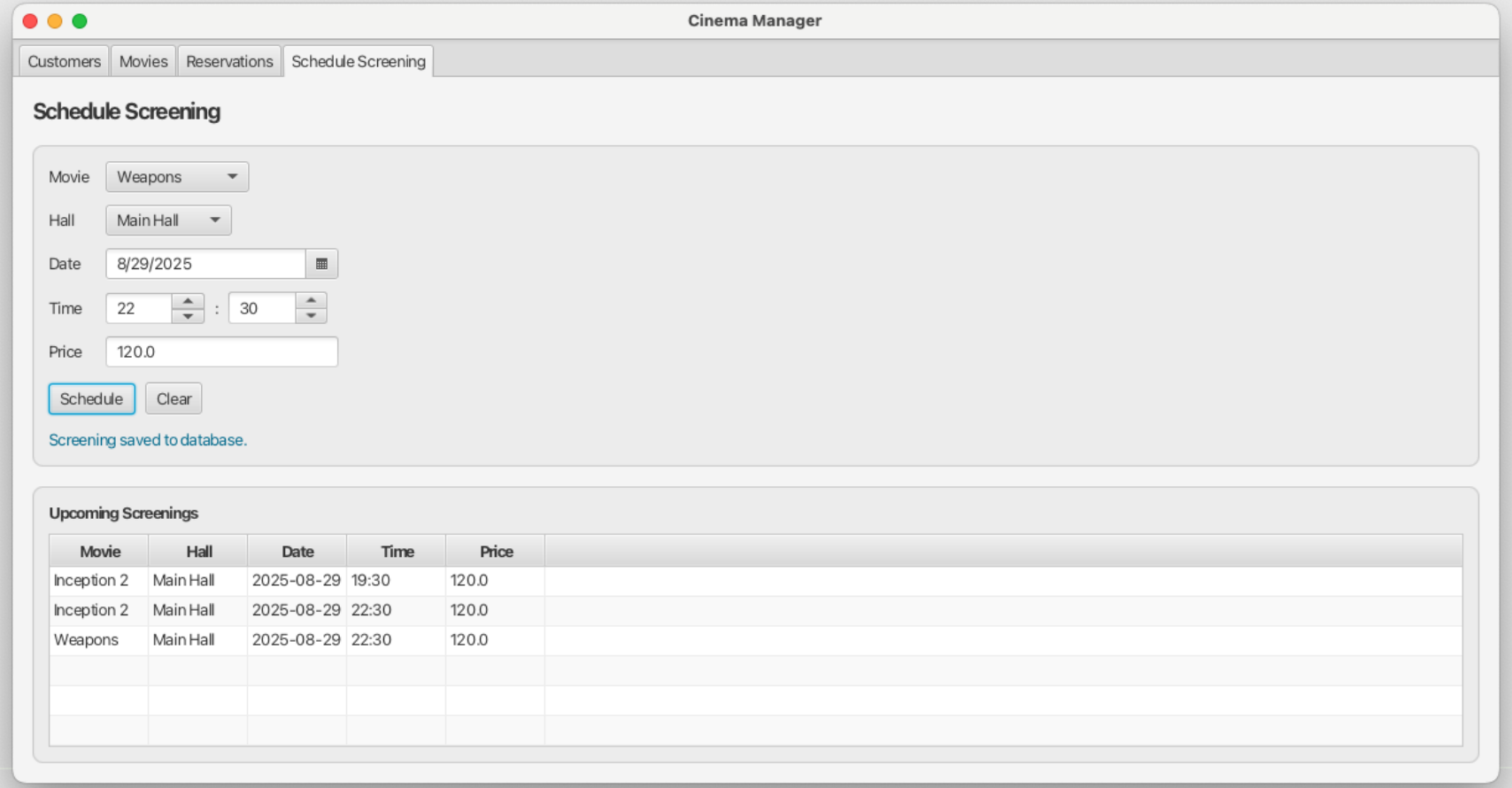
# Movies view


Cinema Manager
Customers
Movies
Reservations
Schedule Screening
Movies
Search by title...
ID
Title
Genre
Language
1
The Matrix
Sci-Fi
English
2
Inception
Action
English
3
Inception 2
Action
English
4
Stand by Me
Drama
English
5
Weapons
Dark Comedy
English
Title:
Weapons
Description:
When all but one child from one class ....
Genre:
Dark Comedy
Language:
English
Duration (min):
128
Release Year:
2025
Add
Update
Delete

# Reservations view



# Schedule screening view



# Other features

